

TEST REPORT

Applicant : Shenzhen Qianhai Linghangwan Technology Co., Ltd.

Address : 18th Floor, Building 1, COFCO Chuangxin R&D Center, Liuxian 1st Road, Xinan Street, Bao'an District, Shenzhen City

Manufacturer : SHENZHEN MAKERFIRE TECHNOLOGY CO.,LTD.

Address : 18th Floor, Building 1, COFCO Chuangxin R&D Center, Liuxian 1st Road, Xinan Street, Bao'an District, Shenzhen City

Product Name : ESP32-S3 Mini Module

Model Number : W11-V0.2

Trademark :  MESHnology

FCC ID : 2BRTY-W11-V02

Test Date : Apr. 01, 2026 -Jun. 29, 2026

Date of Report : Jul. 24, 2026

Test Result : This device described above has been tested by BKC, and the test results show that the equipment under test (EUT) is in compliance with the Standard requirements. And it is applicable only to the tested sample identified in the report.

Test Procedure Used:

Standards : FCC CFR Title 47 Part 15 Subpart C Section 15.247
ANSI C63.10-2020+Cor 1-2023

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1. General Information

1.1. Description of Device (EUT)

EUT Name : ESP32-S3 Mini Module
Model No. : W11-V0.2
Model Difference : /
Power supply : Input: 5VDC 2A
Operation frequency : 2.412-2.462GHz
Max. RF output power: : 22.91 dBm
Modulation : 802.11b: Direct Sequence Spread Spectrum(DSSS)
802.11g/802.11n(H20)/802.11n(H40):
Orthogonal Frequency Division Multiplexing(OFDM)
Antenna Type : FPC antenna, Maximum Gain is 4.07dBi
Hardware Version : V0.2
Software Version : V1.0
Intend use environment : Residential, commercial and light industrial environment

1.2. Channel List

Channel No.	Frequency (MHz)	Channel No.	Frequency (MHz)	Channel No.	Frequency (MHz)	Channel No.	Frequency (MHz)
1	2412	2	2417	3	2422	4	2427
5	2432	6	2437	7	2442	8	2447
9	2452	10	2457	11	2462	/	/
/	/						

1.3. Summary of test result

RF PART		
Test Items	Test Requirement	Result
Antenna requirement	FCC part 15.203/15.247(b)(4)	PASS
AC Power Line Conducted Emission	FCC part 15.207	PASS
Conducted Peak Output Power	FCC part 15.247 (b)(3)	PASS
Channel Bandwidth& 99% OCB	FCC part 15.247 (a)(2)	PASS
Power Spectral Density	FCC part 15.247 (e)	PASS
Band Edge	FCC part 15.247(d)	PASS
Spurious Emission	FCC part 15.205/15.209	PASS
Remark: N/A is an abbreviation for Not Applicable. Measurement Uncertainty is not taken into account for conformity statement		

1.4.Special Accessories and Auxiliary Equipment

Description	Manufacturer	Model No.	Serial Number
Adapter	Salcomp	A2244	F1625031EBDPMG9DK
/	/	/	/

1.5.Test mode

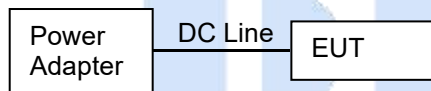
The special RF test software was used to control EUT work in Continuous WIFI TX mode, and select test channel, wireless mode.

Test mode	Description	Low channel	Middle channel	High channel
TM1	802.11b	2412MHz	2437MHz	2462MHz
TM2	802.11g	2412MHz	2437MHz	2462MHz
TM3	802.11n20	2412MHz	2437MHz	2462MHz
TM4	802.11n40	2422MHz	2437MHz	2452MHz

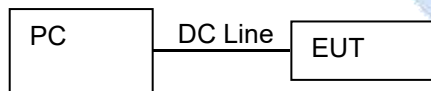
Note: /

1.6.Block Diagram

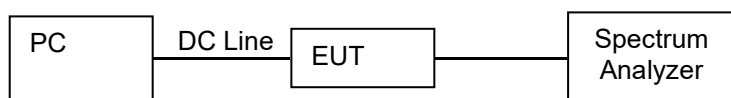
Conducted Emission



Radiated Emission



RF Conducted Test



1.7. Test Conditions

Category	Normal Conditions	Extreme Conditions
Temperature range	15-35°C	N/A
Humidity range	20-75%	N/A
Pressure range	86-106kPa	N/A
Power supply	120Vac for adapter or DC 5V	N/A
Note: N/A is an abbreviation for Not Applicable.		

1.8. Measurement Uncertainty (95% confidence levels, k=2)

Item	MU	Remark
Uncertainty for radio frequency	$\pm 7 \times 10^{-8}$	
Conducted Emissions (150kHz~30MHz)	$\pm 2.54\text{dB}$	
Radiated Emission(9KHz~30MHz)	$\pm 1.6\text{ dB}$	
Radiated Emission(30MHz~1GHz)	Vertical: $\pm 4.14\text{dB}$ Horizontal: $\pm 4.25\text{dB}$	
Radiated Emission(1GHz~6GHz)	$\pm 3.9\text{dB}$	
Radiated Emission(6GHz~18GHz)	$\pm 4.3\text{dB}$	
Radiated Emission(18GHz~40GHz)	$\pm 3.38\text{dB}$	
Uncertainty for conducted RF Power	$\pm 0.20\text{dB}$	
Conducted spurious emissions	$\pm 0.24\text{dB}$	
Uncertainty for temperature	0.6°C	
Uncertainty for humidity	1%	
Bandwidth	$\pm 1.5 \times 10^{-6}$	
Time	$\pm 2\%$	
Duty Cycle	$\pm 2\%$	
Remark: The coverage Factor (k=2), and measurement Uncertainty for a level of Confidence of 95%		

1.9. Test Facility

Site Description		
Name of Firm	:	Shenzhen BKC Testing Co., Ltd.
Site Location	:	Room 103, 1/F, Huaya Building, Huaya Industrial Park, Longhua Community, Longhua District, Shenzhen, Guangdong, China.
ISED CAB identifier	:	CN0188
ISED Company Number	:	34333
FCC Designation Number	:	CN1416
FCC Test Firm Registration Number	:	701543

1.10.Test Equipment

Conducted Test Site

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	LISN	R&S	ENV216	102762	Mar. 11. 2026	Mar. 10. 2027
2	EMI Test Receiver	R&S	ESCI	101424	Mar. 11. 2026	Mar. 10. 2027
3	Rf cables	HTEC	HCE 2M-CE	N/A	Mar. 11. 2026	Mar. 10. 2027
4	Coupling/ Decoupling Network	Diamond	CX210	N/A	Mar. 11. 2026	Mar. 10. 2027
5	CE Testing software	EZ	EZ-EMC	/	/	/

Radiated Test Site(Below 1G and Above 1G)

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Antenna	Schwarzbeck	VULB 9168	01321	Mar. 20. 2025	Mar. 19. 2028
2	EMI Test Receiver	R&S	ESRP	101478	Mar. 11. 2026	Mar. 10. 2027
3	Preamplifier	HP	8447D	2727A05345	Mar. 11. 2026	Mar. 10. 2027
4	Double Ridged Broadband	SCHWARZBECK	BBHA 9120D	02621	Mar. 10. 2025	Mar. 19. 2028
5	Preamplifier	Schwarzbeck	BBV 9718D	00041	Mar. 11. 2026	Mar. 10. 2027
6	Rf cables	HUBER+SUHNER	8M-RE	N/A	Mar. 11. 2026	Mar. 10. 2027
7	Rf cables	HUBER+SUHNER	1.5M-RE	N/A	Mar. 11. 2026	Mar. 10. 2027
8	Rf cables	HUBER+SUHNER	1.5M-AP-RE	N/A	Mar. 09. 2026	Mar. 10. 2027
9	Spectrum	Agilent	N9020A	MY50510311	Mar. 11. 2026	Mar. 10. 2027
10	Spectrum analyzer	R&S	FSP40	200061	Mar. 11. 2026	Mar. 10. 2027
11	Double Ridged Broadband Horn Antenna	COM-POWER	AH-1840	10100009	Mar. 11. 2026	Mar. 10. 2027
12	RE Testing software	EZ	EZ-EMC	/	/	/

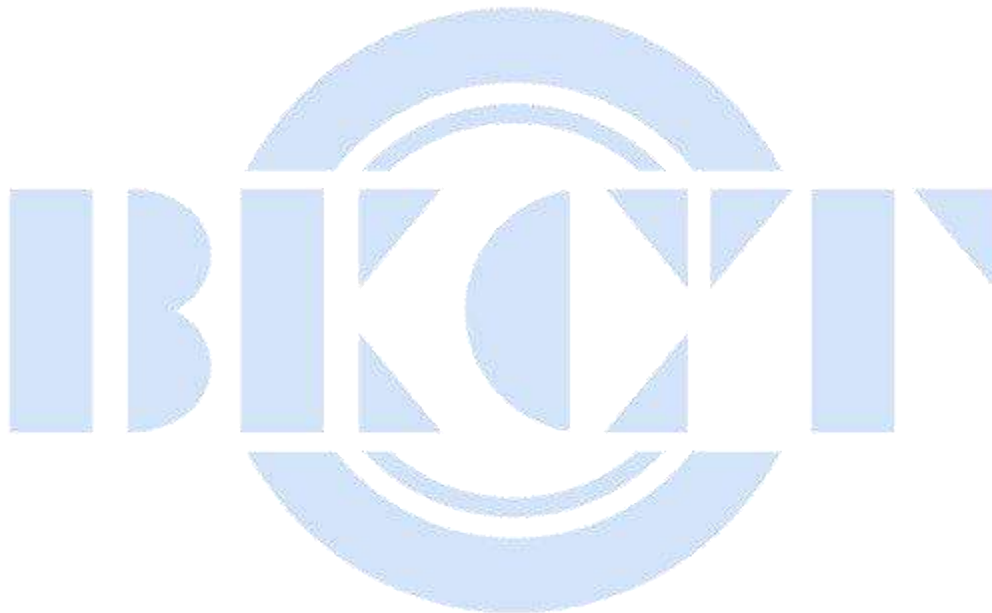
RF Test Site

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Spectrum	Agilent	N9020A	MY50510311	Mar. 11. 2026	Mar. 10. 2027
2	Vector source	Agilent	N5182A	MY47420223	Mar. 11. 2026	Mar. 10. 2027
3	All instrument	Rohde & Schwarz	CMW 500	101131	Mar. 11. 2026	Mar. 10. 2027
4	control unit	MW	MW100-RFCB	MW200411BK-34	Mar. 11. 2026	Mar. 10. 2027
5	Spectrum analyzer	R&S	FSP40	200061	Mar. 11. 2026	Mar. 10. 2027
6	RF Testing software	MW	MTS 8310	/	/	/

Test software

Software name	RF Testing software
Software version	2.0.0.0

Software name	RE Testing software
Software version	EMEC-03A2 RE+
Software name	CE Testing software
Software version	EMEC-3A1+



2. Conducted emission at the mains terminals test

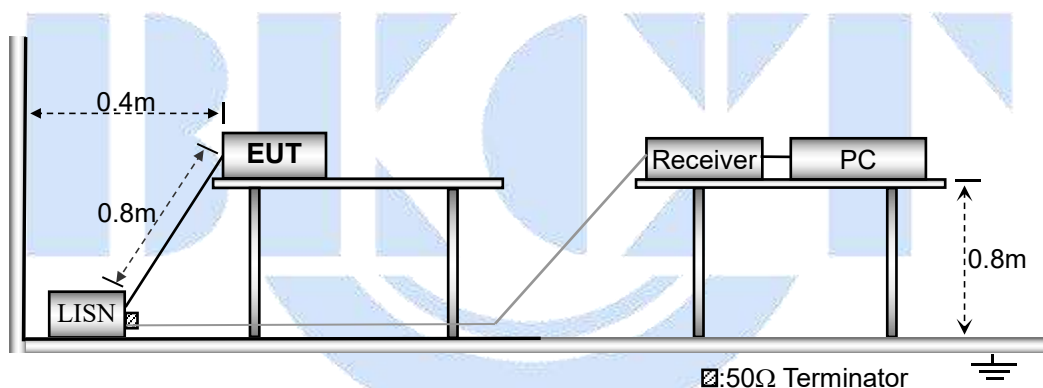
2.1.Limit

Test Requirement:	FCC Part15 C Section 15.207
Test Method:	ANSI C63.10-2020+Cor 1-2023
Test Frequency Range:	150KHz to 30MHz
Receiver setup:	RBW=9KHz, VBW=30KHz, Sweep time=auto

Frequency MHz	Limits dB(μ V)	
	Quasi-peak Level	Average Level
0.15 ~ 0.50	66 ~ 56*	56 ~ 46*
0.50 ~ 5.00	56	46
5.00 ~ 30.00	60	50

Notes: 1. *Decreasing linearly with logarithm of frequency.
2. The lower limit shall apply at the transition frequencies.

2.2.Test Setup



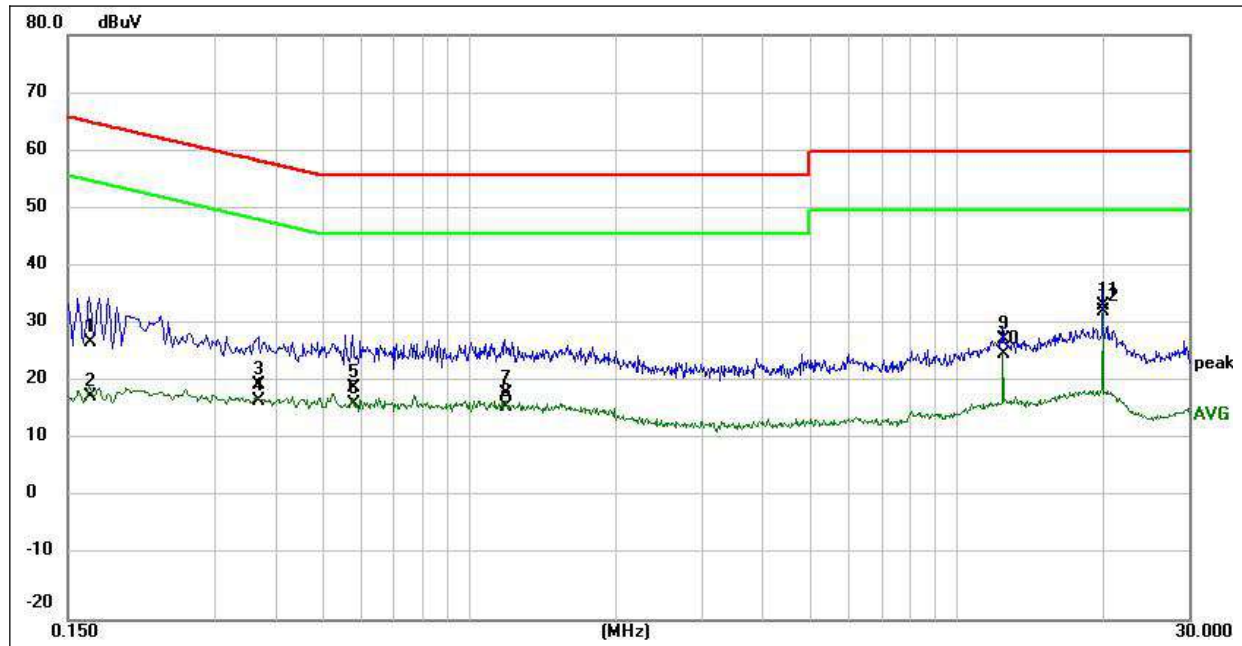
2.3.Test Procedure

1. The EUT was placed 0.8 meters from the horizontal ground plane with EUT being connected to the power mains through a line impedance stabilization network (LISN). All other support equipments powered from additional LISN(s). The LISN provide 50 Ohm/ 50uH of coupling impedance for the measuring instrument.
2. Interconnecting cables that hang closer than 40 cm to the ground plane shall be folded back and forth in the center forming a bundle 30 to 40 cm long.
3. I/O cables that are not connected to a peripheral shall be bundled in the center. The end of the cable may be terminated, if required, using the correct terminating impedance. The overall length shall not exceed 1 m.
4. LISN at least 80 cm from nearest part of EUT chassis.
5. For the actual test configuration, please refer to the related Item - EUT Test Photos.

2.4. Test Result

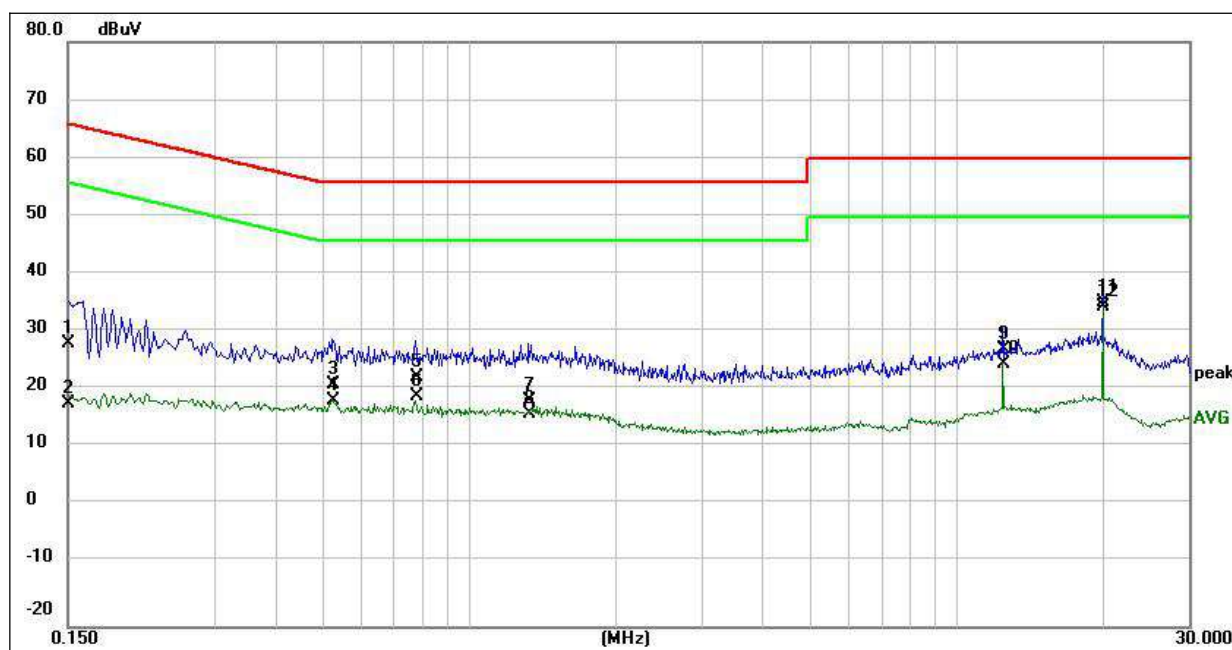
Note: Only the worst Data 802.11b/2412MHz Record in the report.

Conducted Emission At The Mains Terminals Test Data			
Temperature:	25.1 °C	Relative Humidity:	56%
Pressure:	1008hPa	Phase :	Line
Test Voltage :	AC 120V/60Hz for adapter	Test Mode:	TM1-2412MHz
Model No. :	W11-V0.2		



No.	Frequency (MHz)	Reading Level(dBuV)	Factor (dB)	Measurement(dBuV)	Limit (dBuV)	Margin (dB)	Detector
1	0.1660	6.80	20.61	27.41	65.16	-37.75	QP
2	0.1660	-2.37	20.61	18.24	55.16	-36.92	AVG
3	0.3700	-1.59	21.89	20.30	58.50	-38.20	QP
4	0.3700	-4.40	21.89	17.49	48.50	-31.01	AVG
5	0.5780	-2.17	21.89	19.72	56.00	-36.28	QP
6	0.5780	-4.91	21.89	16.98	46.00	-29.02	AVG
7	1.1860	-2.95	21.69	18.74	56.00	-37.26	QP
8	1.1860	-5.24	21.69	16.45	46.00	-29.55	AVG
9	12.5020	8.04	19.92	27.96	60.00	-32.04	QP
10	12.5020	5.66	19.92	25.58	50.00	-24.42	AVG
11	20.0020	13.82	19.99	33.81	60.00	-26.19	QP
12	20.0020	12.73	19.99	32.72	50.00	-17.28	AVG

Conducted Emission At The Mains Terminals Test Data			
Temperature:	25.1 °C	Relative Humidity:	56%
Pressure:	1008hPa	Phase :	Neutral
Test Voltage :	AC 120V/60Hz for adapter	Test Mode:	TM1-2412MHz
Model No. :	W11-V0.2		



No.	Frequency (MHz)	Reading Level(dBuV)	Factor (dB)	Measure-ment(dBuV)	Limit (dBuV)	Margin (dB)	Detector
1	0.1500	8.50	19.98	28.48	66.00	-37.52	QP
2	0.1500	-1.88	19.98	18.10	56.00	-37.90	AVG
3	0.5260	-0.30	21.89	21.59	56.00	-34.41	QP
4	0.5260	-3.16	21.89	18.73	46.00	-27.27	AVG
5	0.7780	0.78	21.88	22.66	56.00	-33.34	QP
6	0.7780	-2.35	21.88	19.53	46.00	-26.47	AVG
7	1.3260	-2.81	21.55	18.74	56.00	-37.26	QP
8	1.3260	-5.20	21.55	16.35	46.00	-29.65	AVG
9	12.5020	7.60	19.92	27.52	60.00	-32.48	QP
10	12.5020	4.98	19.92	24.90	50.00	-25.10	AVG
11	20.0020	15.56	19.99	35.55	60.00	-24.45	QP
12	20.0020	14.80	19.99	34.79	50.00	-15.21	AVG

3. Radiated Spurious Emissions

3.1.Limit

Test Requirement: FCC CFR47 Part 15 Section 15.209 & 15.247
Test Method: KDB 558074 D01 15.247 Meas Guidance v05r02;
ANSI C63.10-2020+Cor 1-2023
Test Result: PASS
Measurement Distance: 3m

Frequency	Detector	RBW	VBW	Value
9KHz-150KHz	Quasi-peak	200Hz	600Hz	Quasi-peak
150KHz-30MHz	Quasi-peak	9KHz	30KHz	Quasi-peak
30MHz-1GHz	Quasi-peak	120KHz	360KHz	Quasi-peak
Above 1GHz	Peak	1MHz	3MHz	Peak
	Peak	1MHz	10Hz	Average

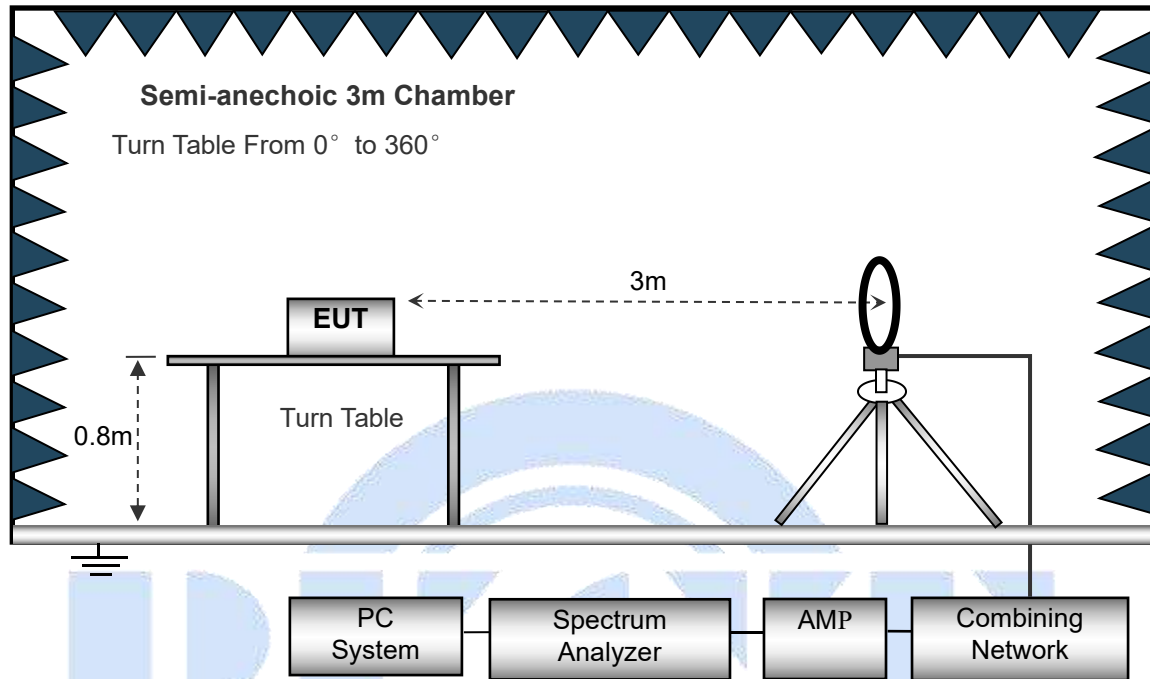
Limit:

Frequency (MHz)	Field Strength		Field Strength Limit at 3m Measurement Dist	
	uV/m	Distance (m)	uV/m	dBuV/m
0.009 ~ 0.490	$2400/F(\text{kHz})$	300	$10000 * 2400/F(\text{kHz})$	$20\log(2400/F(\text{kHz})) + 80$
0.490 ~ 1.705	$24000/F(\text{kHz})$	30	$100 * 24000/F(\text{kHz})$	$20\log(24000/F(\text{kHz})) + 40$
1.705 ~ 30	30	30	$100 * 30$	$20\log(30) + 40$
30 ~ 88	100	3	100	$20\log(100)$
88 ~ 216	150	3	150	$20\log(150)$
216 ~ 960	200	3	200	$20\log(200)$
Above 960	500	3	500	$20\log(500)$

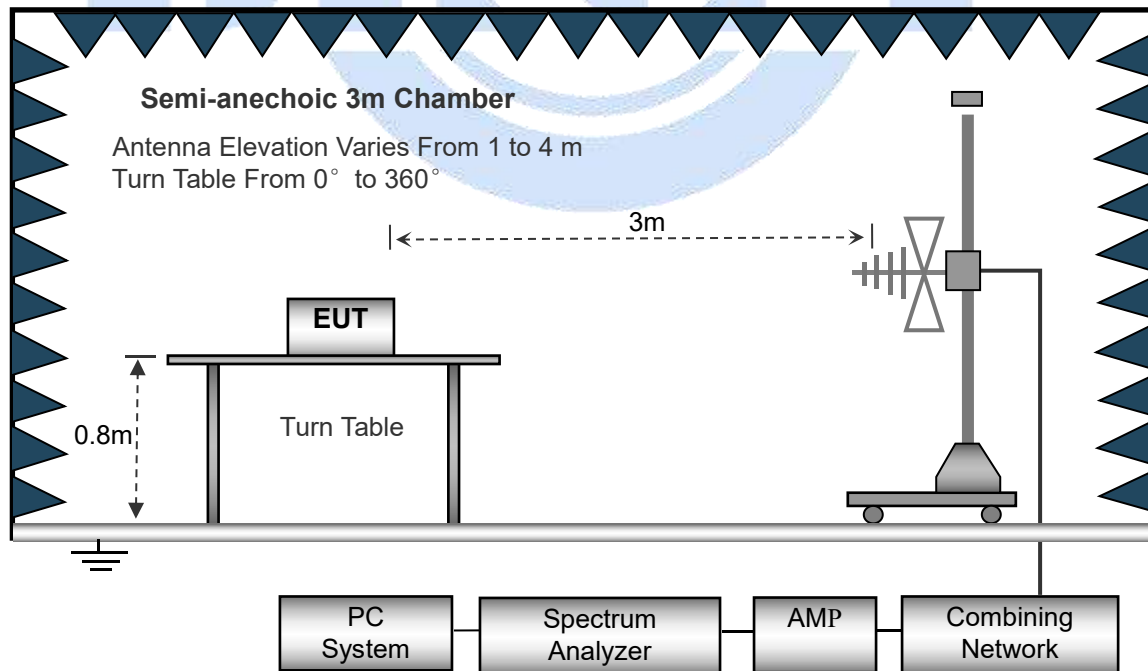
3.2. Test Setup

The radiated emission tests were performed in the 3m Semi- Anechoic Chamber test site, using the setup accordance with the ANSI C63.10.

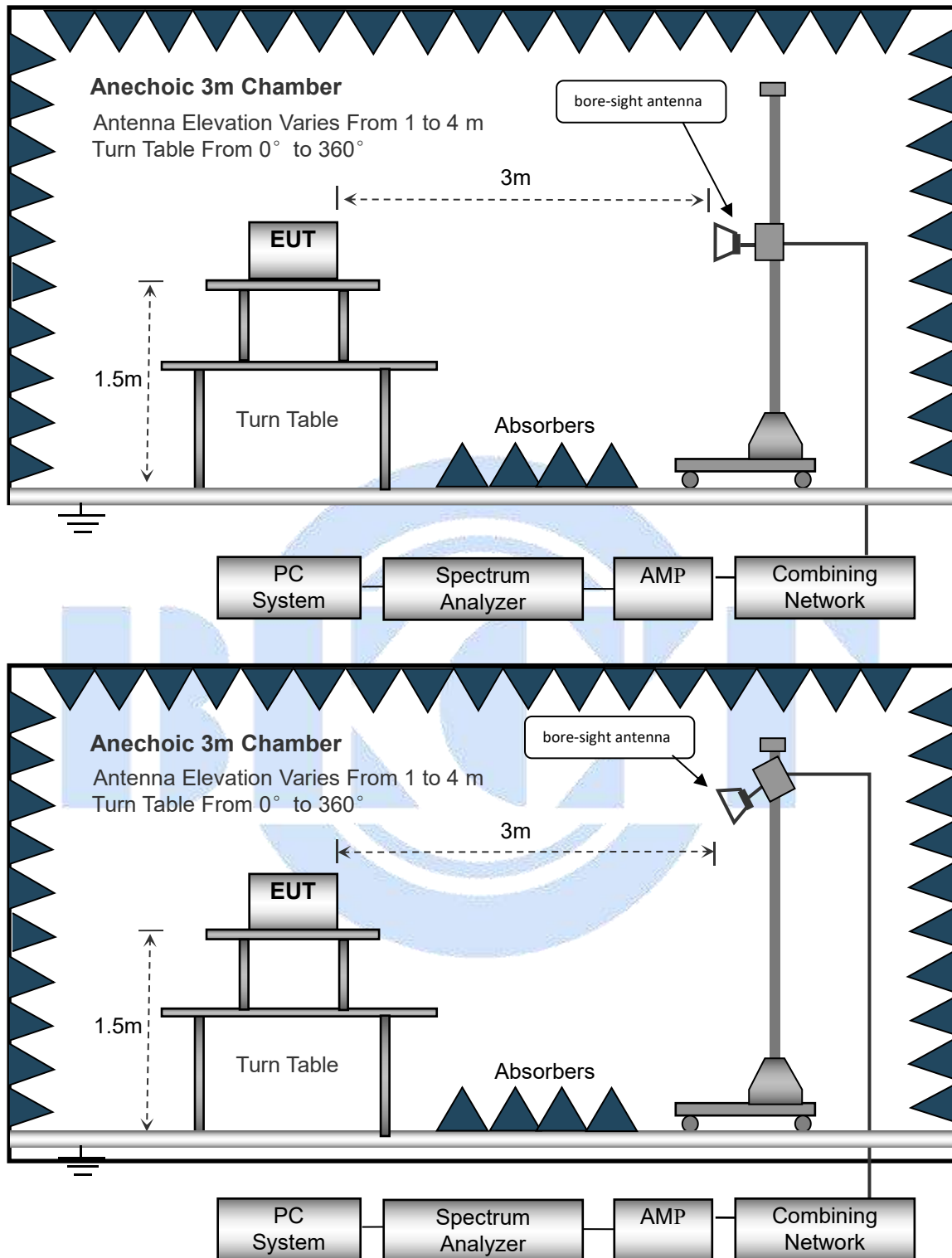
The test setup for emission measurement below 30MHz.



The test setup for emission measurement from 30 MHz to 1 GHz.

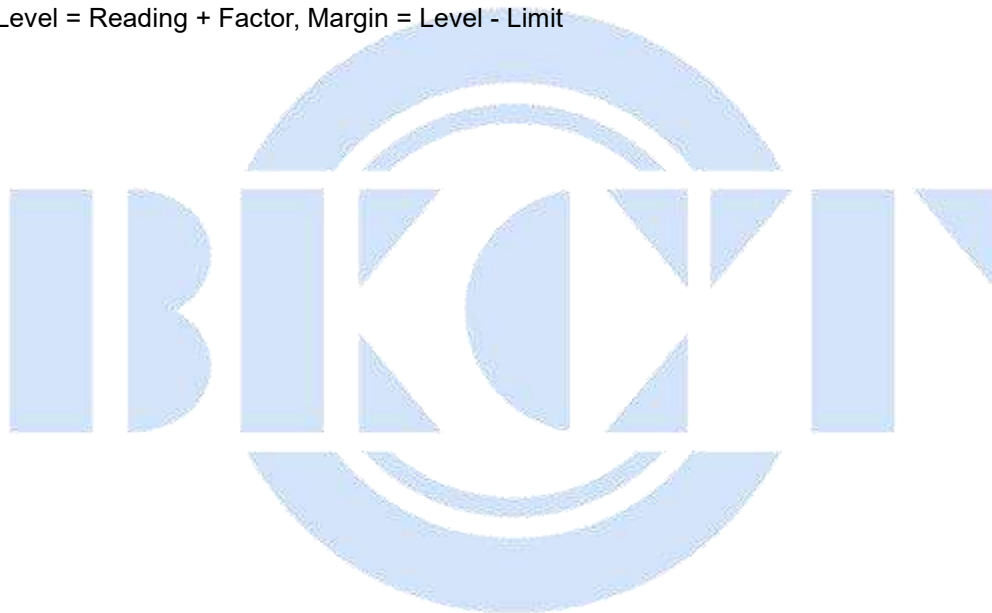


The test setup for emission measurement above 1 GHz.



3.3. Test Procedure

1. The EUT is placed on a turntable, which is 0.8m above ground plane for below 1GHz and 1.5m for above 1GHz.
2. The turntable shall be rotated for 360 degrees to determine the position of maximum emission level.
3. EUT is set 3m away from the receiving antenna, which is moved from 1m to 4m to find out the maximum emissions.
4. Maximum procedure was performed on the six highest emissions to ensure EUT compliance.
5. And also, each emission was to be maximized by changing the polarization of receiving antenna both horizontal and vertical.
6. Repeat above procedures until the measurements for all frequencies are complete.
7. The radiation measurements are performed in X,Y and Z axis positioning(X denotes lying on the table, Y denotes side stand and Z denotes vertical stand),the worst condition was tested putting the eut in Z axis,so the worst data were shown as follow.
8. A 2.4GHz high - pass filter is used during radiated emissions above 1GHz measurement.
9. Remark: Level = Reading + Factor, Margin = Level - Limit



3.4. Test Result

Test Frequency: 9KHz~30MHz

The measurements were more than 20 dB below the limit and not reported.

Note: Only the worst Data 802.11b/2412MHz Record in the report.

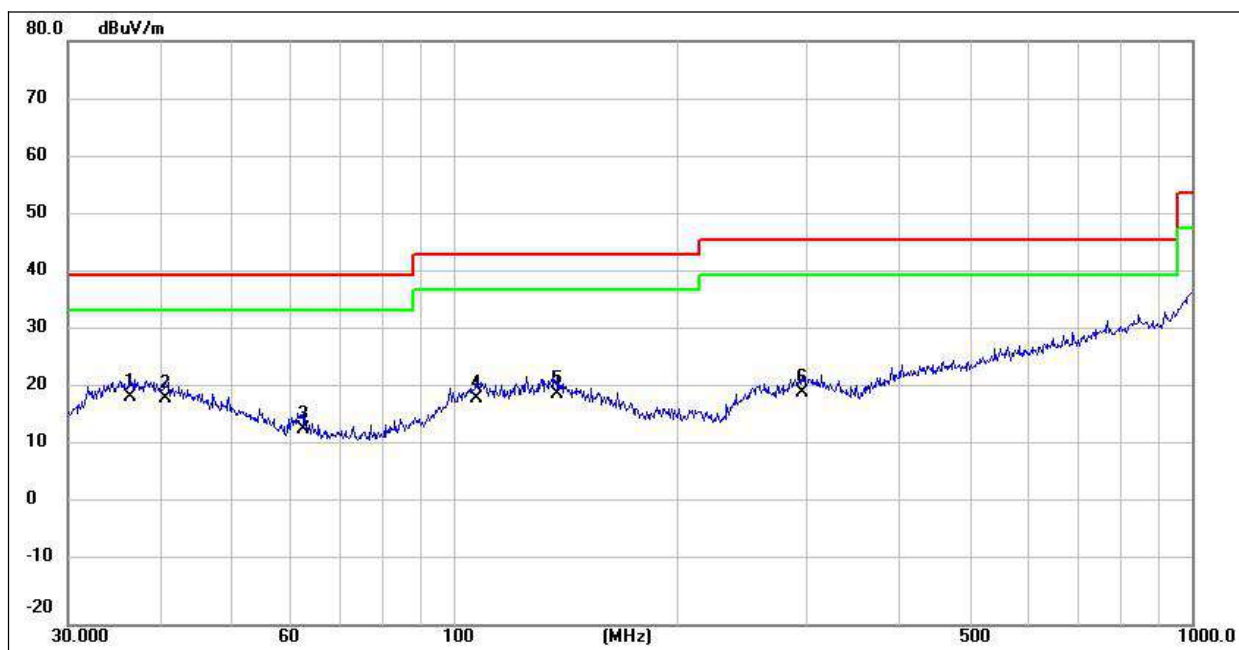
Test Frequency : 30MHz ~ 1GHz

Radiation Emission Test Data			
Temperature:	25.1℃	Relative Humidity:	56%
Pressure:	1008hPa	Phase :	Horizontal
Test Voltage :	AC 120V/60Hz for adapter	Test Mode	TM1-2412MHz
Model No. :	W11-V0.2		



No.	Frequency (MHz)	Reading Level (dBuV)	Correct Factor (dB/m)	Emission Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Remark			
1	38.0783	21.10	-2.04	19.06	40.00	-20.94	QP			
2	61.9951	22.47	-9.93	12.54	40.00	-27.46	QP			
3	102.0014	24.28	-4.66	19.62	43.50	-23.88	QP			
4	135.5062	21.73	-2.89	18.84	43.50	-24.66	QP			
5	261.9753	20.17	-2.38	17.79	46.00	-28.21	QP			
6	314.3765	21.45	-1.86	19.59	46.00	-26.41	QP			

Radiation Emission Test Data			
Temperature:	25.1°C	Relative Humidity:	56%
Pressure:	1008hPa	Phase :	Vertical
Test Voltage :	AC 120V/60Hz for adapter	Test Mode	TM1-2412MHz
Model No. :	W11-V0.2		



No.	Frequency (MHz)	Reading Level (dBuV)	Correct Factor (dB/m)	Emission Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Remark			
1	36.5092	21.14	-2.05	19.09	40.00	-20.91	QP			
2	40.7016	21.09	-2.29	18.80	40.00	-21.20	QP			
3	62.4313	23.60	-9.96	13.64	40.00	-26.36	QP			
4	107.1337	23.08	-4.34	18.74	43.50	-24.76	QP			
5	138.3873	22.22	-2.69	19.53	43.50	-23.97	QP			
6	296.1836	20.96	-1.15	19.81	46.00	-26.19	QP			

Test Frequency 1GHz-25GHz:

All modes 2.4G 802.11b/g/n have been tested, and the worst result 802.11b_2412MHz recorded was report as below:

Test mode: 802.11b		Frequency:		Channel 1: 2412MHz			
Freq. (MHz)	Ant.Pol.	Emission Level(dBuV/m)		Limit 3m(dBuV/m)		Over(dB)	
	H/V	PK	AV	PK	AV	PK	AV
4824	V	51.55	43.51	74	54	-22.45	-10.49
7236	V	57.29	40.44	74	54	-16.71	-13.56
9648	V	45.91	37.98	74	54	-28.09	-16.02
12060	V	51.71	38.72	74	54	-22.29	-15.28
14472	V	48.02	38.78	74	54	-25.98	-15.22
4824	H	48.83	43.97	74	54	-25.17	-10.03
7236	H	45.08	39.96	74	54	-28.92	-14.04
9648	H	59.38	41.85	74	54	-14.62	-12.15
12060	H	50.16	43.09	74	54	-23.84	-10.91
14472	H	45.97	35.45	74	54	-28.03	-18.55

Note: (1) All Readings are Peak Value (VBW=3MHz) and Peak Value (VBW=10Hz).
(2) Emission Level= Reading Level+Probe Factor +Cable Loss.

Radiated Band Emission Measurement:

All modes 2.4G 802.11b/g/n have been tested, and the worst result 802.11b recorded was report as below:

Test mode: 802.11b		Frequency:		Channel 1: 2412MHz	
Frequency (MHz)	Polarity H/V	PK(dBuV/m) (VBW=3MHz)	Limit 3m (dBuV/m)	AV(dBuV/m) (VBW=10Hz)	Limit 3m (dBuV/m)
2310.00	H	46.19	74.00	43.24	54.00
2390.00	H	54.39	74.00	37.05	54.00
2310.00	V	52.90	74.00	43.00	54.00
2390.00	V	51.46	74.00	43.47	54.00

Test mode: 802.11b		Frequency:		Channel 11: 2462MHz	
Frequency (MHz)	Polarity H/V	PK(dBuV/m) (VBW=3MHz)	Limit 3m (dBuV/m)	AV(dBuV/m) (VBW=10Hz)	Limit 3m (dBuV/m)
2483.50	H	45.77	74.00	36.71	54.00
2500.00	H	59.81	74.00	40.36	54.00
2483.50	V	46.75	74.00	39.97	54.00
2500.00	V	51.83	74.00	36.93	54.00

Note: (1) All Readings are Peak Value (VBW=3MHz) and Peak Value (VBW=10Hz)
(2) Emission Level= Reading Level+Probe Factor +Cable Loss.

4. Conducted Spurious Emissions

4.1.Limit

Test Requirement: FCC CFR47 Part 15 Section 15.247
Test Method: KDB 558074 D01 15.247 Meas Guidance v05r02;
ANSI C63.10-2020+Cor 1-2023
Test Result: PASS

Limit:

In any 100 kHz bandwidth outside the frequency band in which the spread spectrum or digitally modulated intentional radiator is operating, the radio frequency power that is produced by the intentional radiator shall be at least 20 dB below that in the 100 kHz bandwidth within the band that contains the highest level of the desired power, based on either an RF conducted or a radiated measurement, provided the transmitter demonstrates compliance with the peak conducted power limits. If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB. Attenuation below the general limits specified in Section 15.209(a) is not required. In addition, radiated emissions which fall in the restricted bands, as defined in Section 15.205(a), must also comply with the radiated emission limits specified in Section 15.209(a) (see Section 15.205(c)).

4.2.Test Setup



4.3.Test Procedure

- 1) Remove the antenna from the EUT and connect to the spectrum analyzer via a low loss RF cable.
- 2) Set instrument center frequency to DTS channel center frequency.
- 3) Set the span to 1.5 times the DTS bandwidth.
- 4) Set the RBW = 100 kHz.
- 5) Set the VBW [3 × RBW].
- 6) Detector = peak.
- 7) Sweep time = auto couple.
- 8) Trace mode = max hold.
- 9) Allow trace to fully stabilize.
- 10) Use the peak marker function to determine the maximum PSD level.

Emission level measurement:

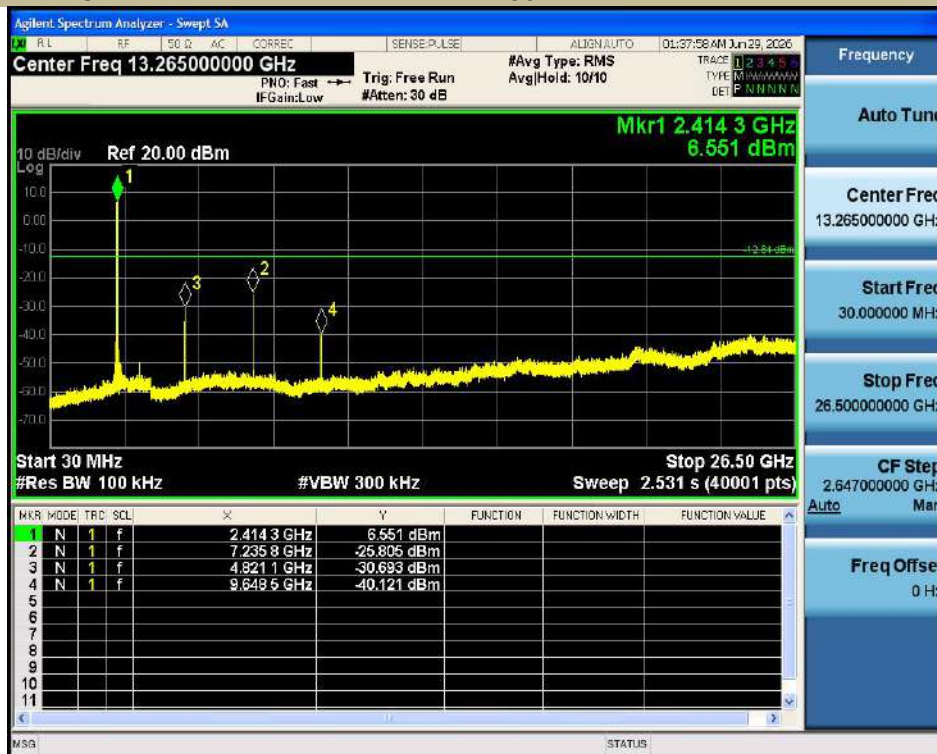
- 1) Set the center frequency and span to encompass frequency range to be measured.
- 2) Set the RBW = 100 kHz.
- 3) Set the VBW [3 × RBW].
- 4) Detector = peak.
- 5) Sweep time = auto couple.
- 6) Trace mode = max hold.
- 7) Allow trace to fully stabilize.
- 8) Use the peak marker function to determine the maximum amplitude level.

4.4.Test Result

Test Model

Maximum Conducted Level RBW=100kHz
Channel 1: 2412MHz
802.11b

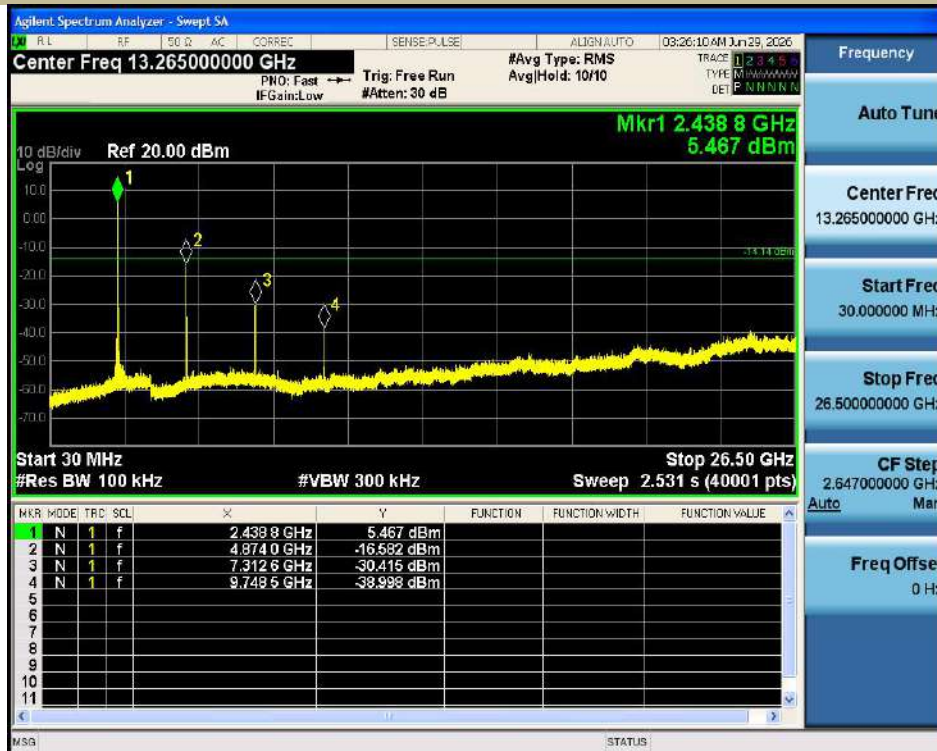

Test Model

Conducted Spurious RF Conducted Emission
Channel 1: 2412MHz
802.11b


Test Model

Maximum Conducted Level RBW=100kHz
Channel 6: 2437MHz
802.11b


Test Model

Conducted Spurious RF Conducted Emission
Channel 6: 2437MHz
802.11b


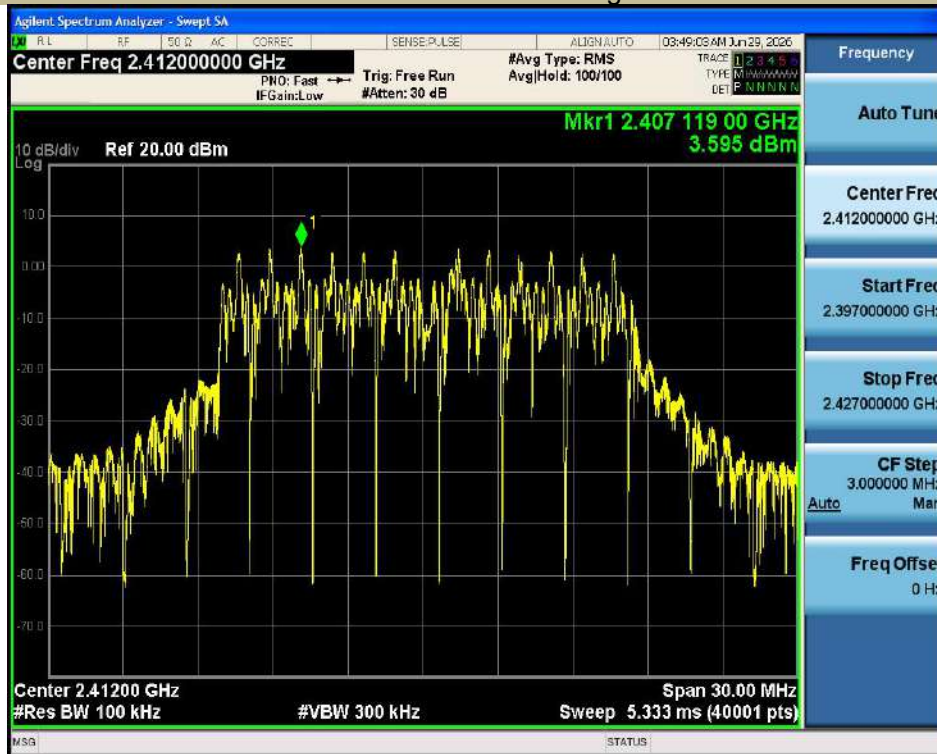
Test Model

Maximum Conducted Level RBW=100kHz
Channel 11: 2462MHz
802.11b


Test Model

Conducted Spurious RF Conducted Emission
Channel 11: 2462MHz
802.11b

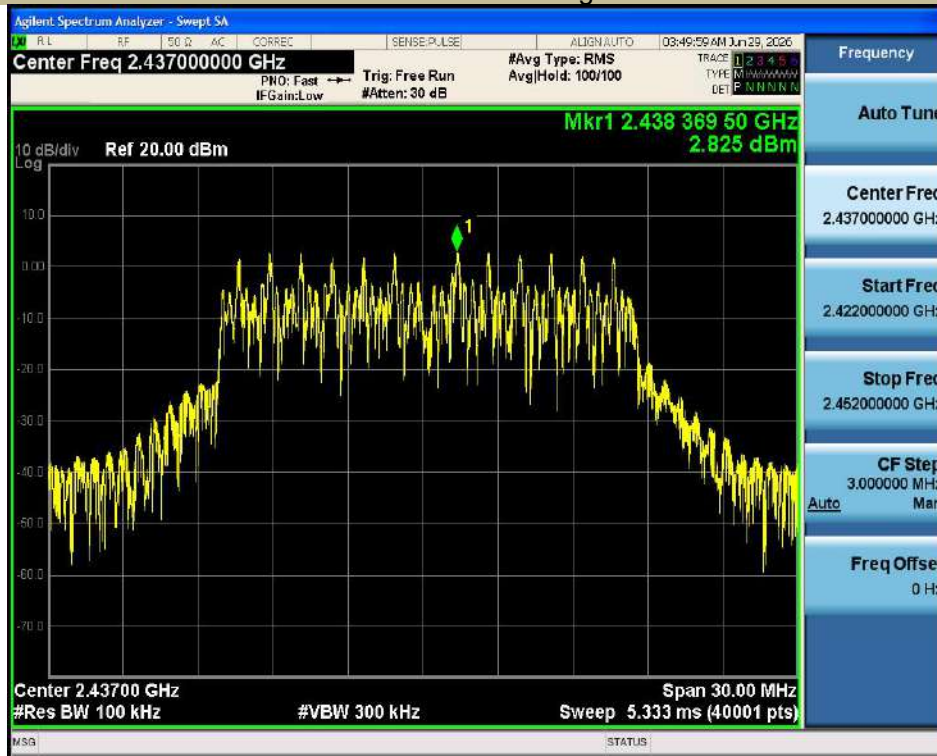

Test Model

Maximum Conducted Level RBW=100kHz
Channel 1: 2412MHz 802.11g


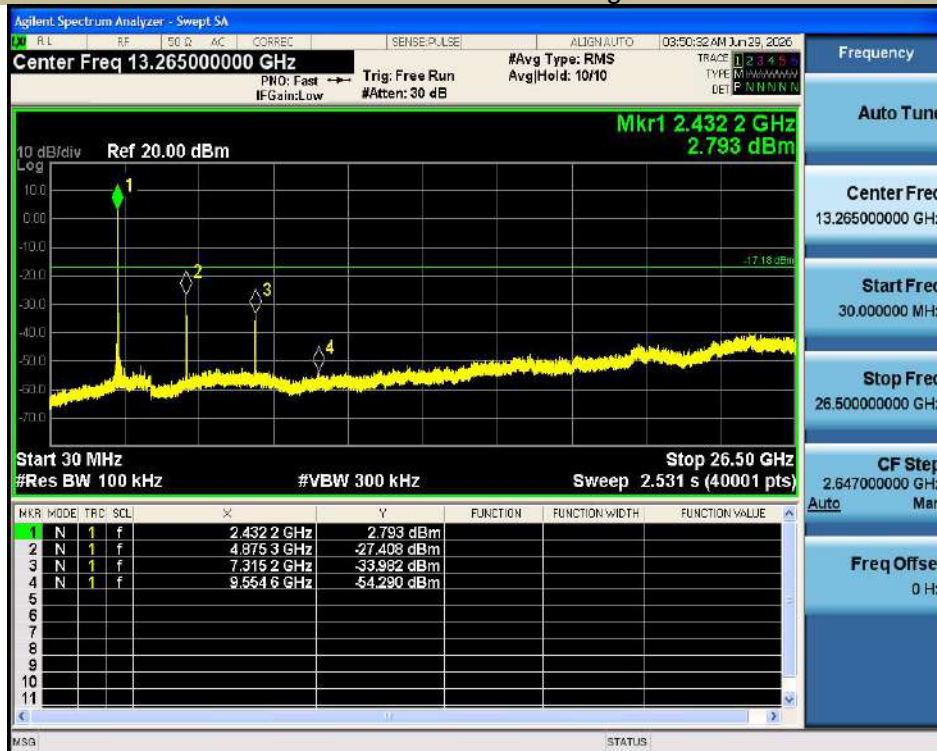
Test Model

Conducted Spurious RF Conducted Emission
Channel 1: 2412MHz 802.11g

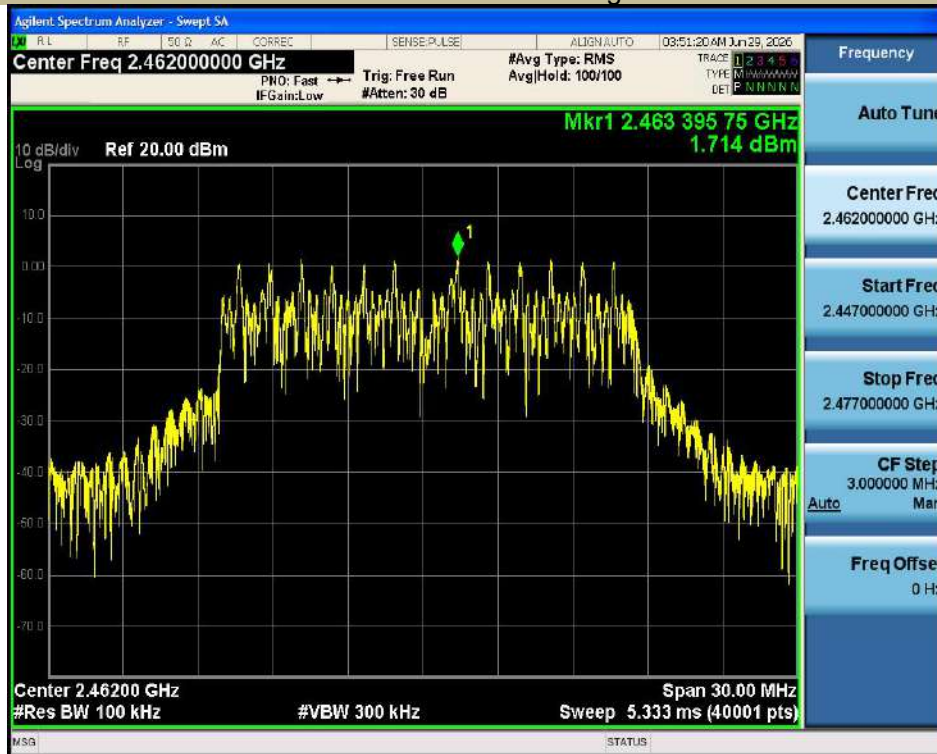

Test Model

Maximum Conducted Level RBW=100kHz
Channel 6: 2437MHz
802.11g


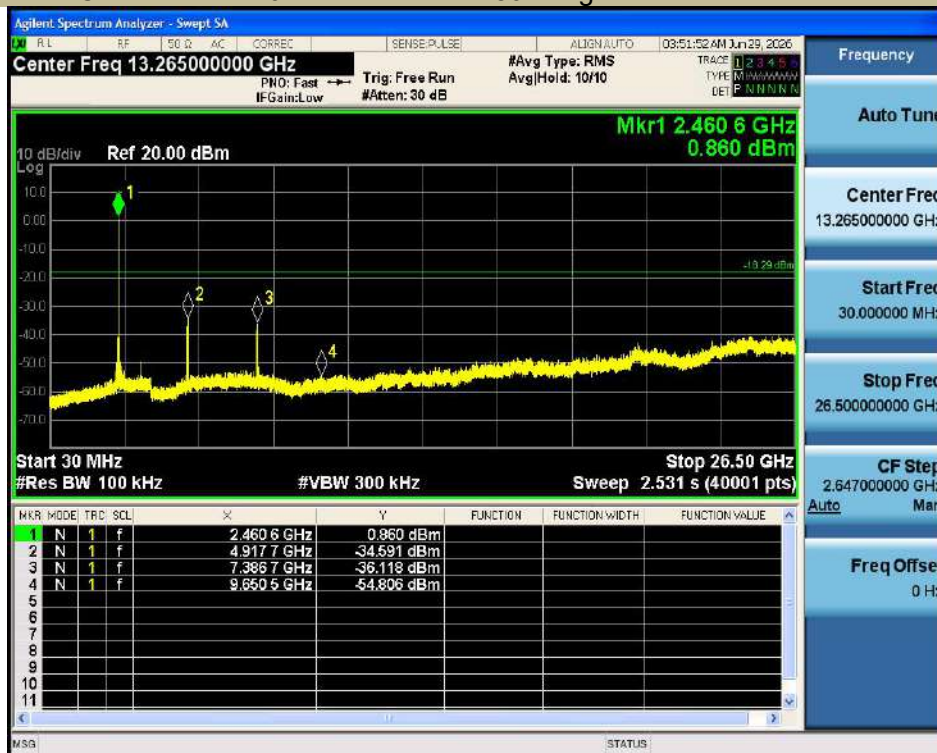
Test Model

Conducted Spurious RF Conducted Emission
Channel 6: 2437MHz
802.11g


Test Model

Maximum Conducted Level RBW=100kHz
Channel 11: 2462MHz 802.11g


Test Model

Conducted Spurious RF Conducted Emission
Channel 11: 2462MHz 802.11g


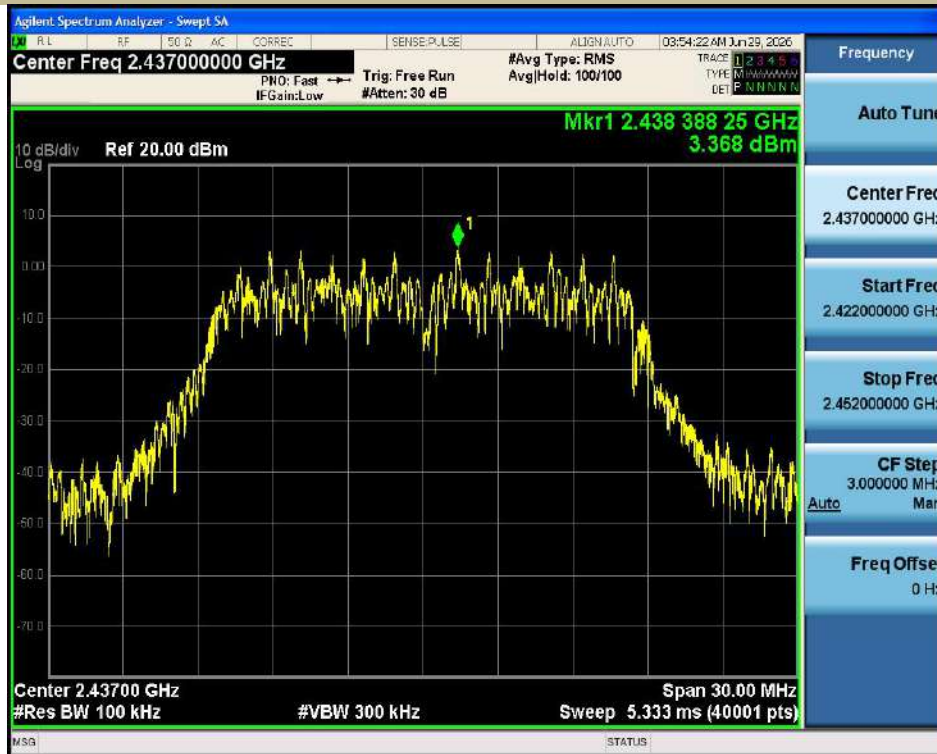
Test Model

Maximum Conducted Level RBW=100kHz
Channel 1: 2412MHz 802.11n20


Test Model

Conducted Spurious RF Conducted Emission
Channel 1: 2412MHz 802.11n20


Test Model

Maximum Conducted Level RBW=100kHz
Channel 6: 2437MHz
802.11n20


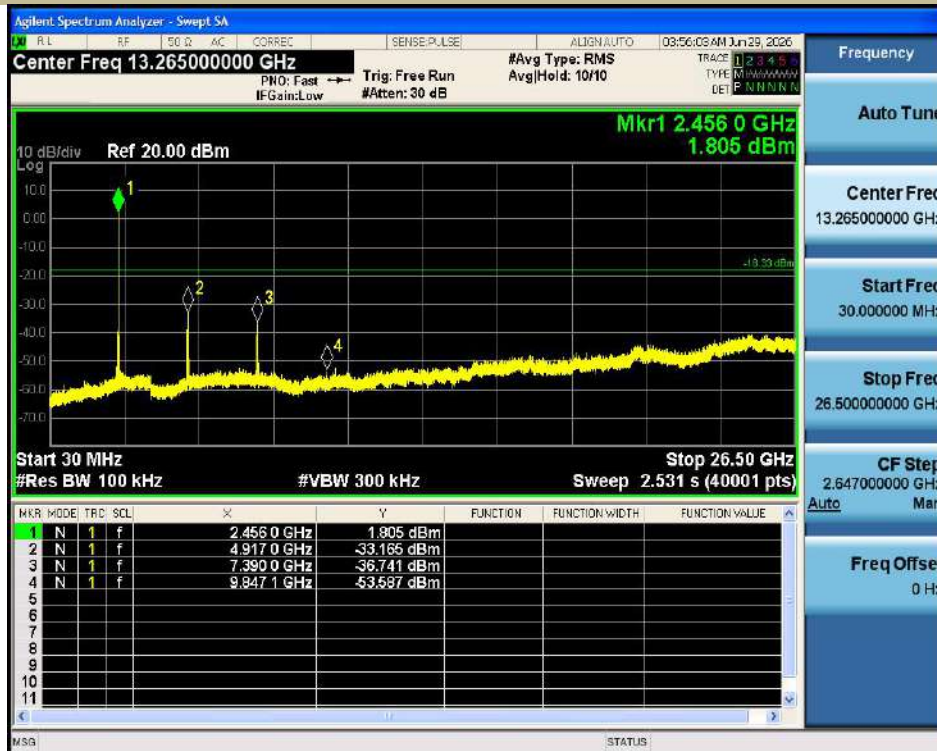
Test Model

Conducted Spurious RF Conducted Emission
Channel 6: 2437MHz
802.11n20


Test Model

Maximum Conducted Level RBW=100kHz
Channel 11: 2462MHz 802.11n20

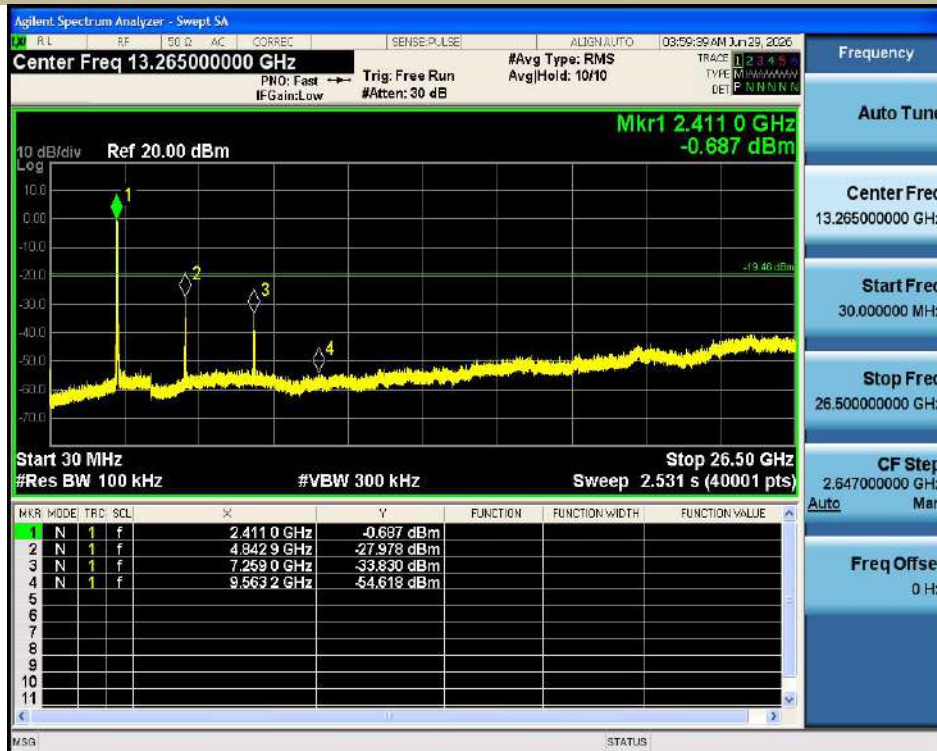

Test Model

Conducted Spurious RF Conducted Emission
Channel 11: 2462MHz 802.11n20


Test Model

Maximum Conducted Level RBW=100kHz
Channel 3: 2422MHz 802.11n40

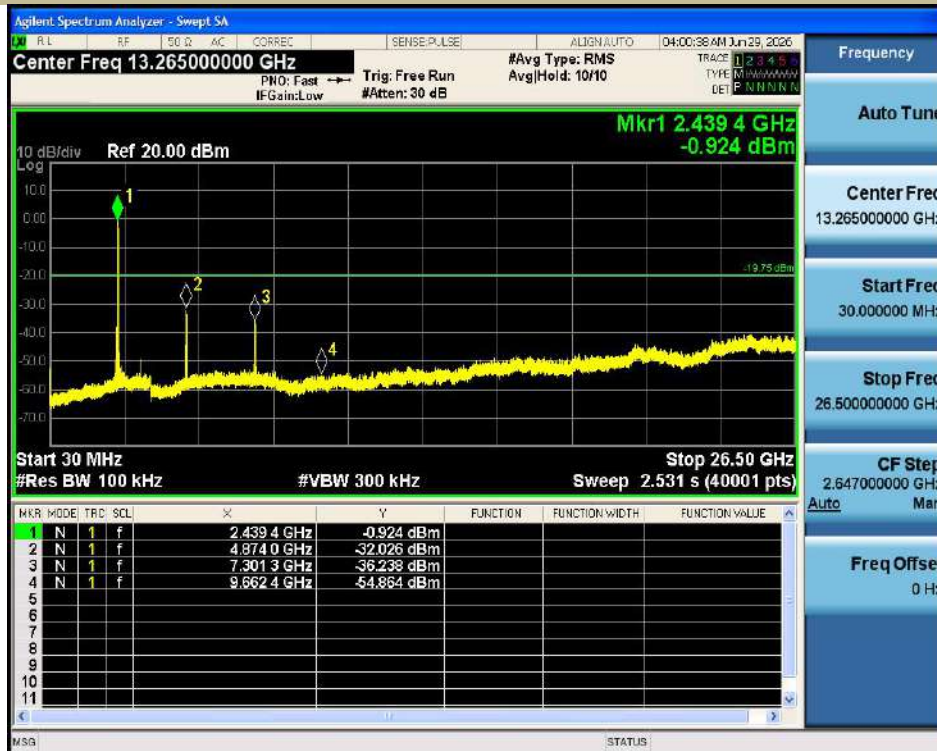

Test Model

Conducted Spurious RF Conducted Emission
Channel 3: 2422MHz 802.11n40


Test Model

Maximum Conducted Level RBW=100kHz
Channel 6: 2437MHz
802.11n40


Test Model

Conducted Spurious RF Conducted Emission
Channel 6: 2437MHz
802.11n40


Test Model

Maximum Conducted Level RBW=100kHz
Channel 9: 2452MHz
802.11n40


Test Model

Conducted Spurious RF Conducted Emission
Channel 9: 2452MHz
802.11n40


5. Band Edge Measurement

5.1.Limit

Test Requirement: FCC CFR47 Part 15 Section 15.247

Test Method: KDB 558074 D01 15.247 Meas Guidance v05r02;

ANSI C63.10-2020+Cor 1-2023

Test Limit: Regulation 15.247 (d), In any 100 kHz bandwidth outside the frequency band which the spread spectrum or digitally modulated intentional radiator is operating, the radio frequency power that is produced by the intentional radiator shall be at least 20 dB below that in the 100 kHz bandwidth within the band that contains the highest level of the desired power, based on either an RF conducted or a radiated measurement, provided the transmitter demonstrates compliance with the peak conducted power limits. If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB. Attenuation below the general limits specified in §15.209(a) is not required. In addition, radiated emissions which fall in the restricted bands, as defined in §15.205(a), must also comply with the radiated emission limits specified in §15.209(a) (see §15.205(c)).

Test Mode: TM1

5.2.Test setup



5.3.Test Procedure

- 1) Remove the antenna from the EUT and connect to the spectrum analyzer via a low loss RF cable.
- 2) Set instrument center frequency to DTS channel center frequency.
- 3) Set the span to 1.5 times the DTS bandwidth.
- 4) Set the RBW = 100 kHz.
- 5) Set the VBW [3 × RBW].
- 6) Detector = peak.
- 7) Sweep time = auto couple.
- 8) Trace mode = max hold.
- 9) Allow trace to fully stabilize.
- 10) Use the peak marker function to determine the maximum PSD level.

Emission level measurement:

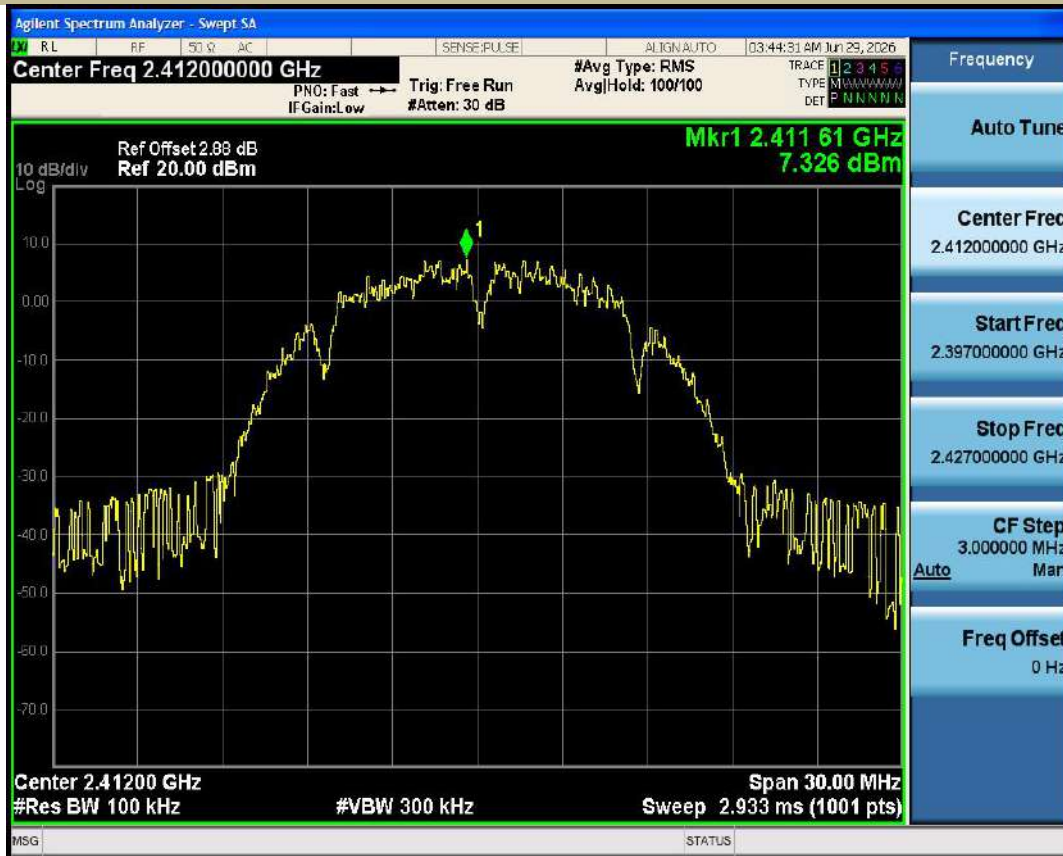
- 1) Set the center frequency and span to encompass frequency range to be measured.
- 2) Set the RBW = 100 kHz.
- 3) Set the VBW [3 × RBW].
- 4) Detector = peak.
- 5) Sweep time = auto couple.
- 6) Trace mode = max hold.
- 7) Allow trace to fully stabilize.
- 8) Use the peak marker function to determine the maximum amplitude level.

5.4.Test Result

Test Model

Band-edge Conducted Emissions
Channel 1: 2412MHz

802.11b



Test Model

Band-edge Conducted Emissions
Channel 1: 2412MHz

802.11b



Test Model

Band-edge Conducted Emissions
Channel 11: 2462MHz

802.11b



Test Model

Band-edge Conducted Emissions
Channel 11: 2462MHz

802.11b



Test Model

Band-edge Conducted Emissions
Channel 1: 2412MHz

802.11g



Test Model

Band-edge Conducted Emissions
Channel 1: 2412MHz

802.11g



Test Model

Band-edge Conducted Emissions
Channel 11: 2462MHz

802.11g



Test Model

Band-edge Conducted Emissions
Channel 11: 2462MHz

802.11g



Test Model

Band-edge Conducted Emissions
Channel 1: 2412MHz

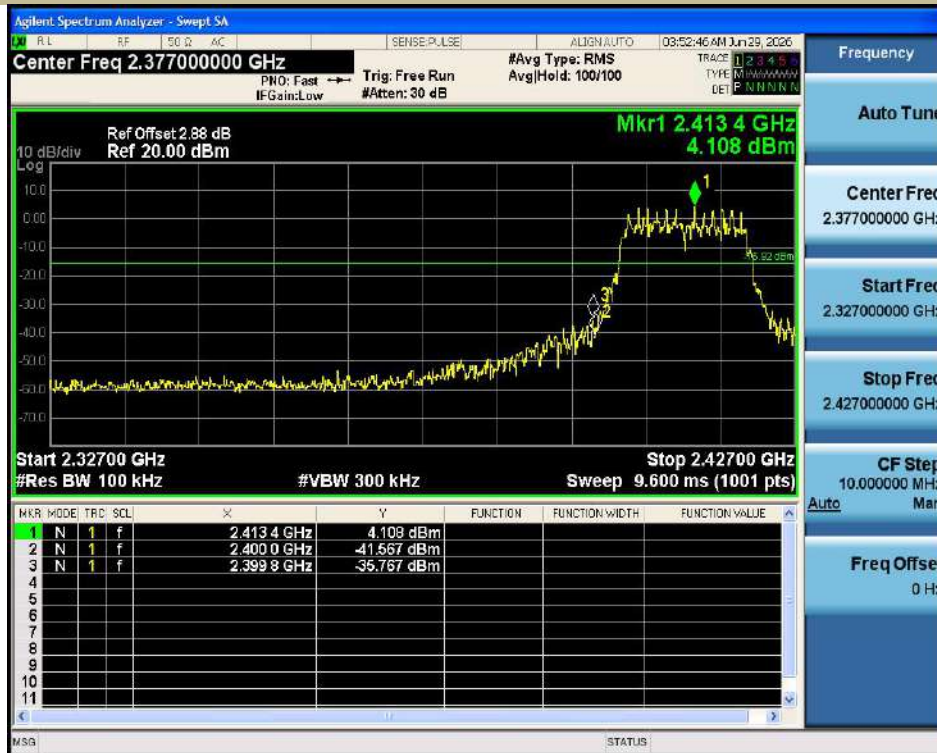
802.11n20



Test Model

Band-edge Conducted Emissions
Channel 1: 2412MHz

802.11n20



Test Model

Band-edge Conducted Emissions
Channel 11: 2462MHz

802.11n20



Test Model

Band-edge Conducted Emissions
Channel 11: 2462MHz

802.11n20



Test Model

Band-edge Conducted Emissions
Channel 3: 2422MHz

802.11n40



Test Model

Band-edge Conducted Emissions
Channel 3: 2422MHz

802.11n40



Test Model

Band-edge Conducted Emissions
Channel 6: 2452MHz

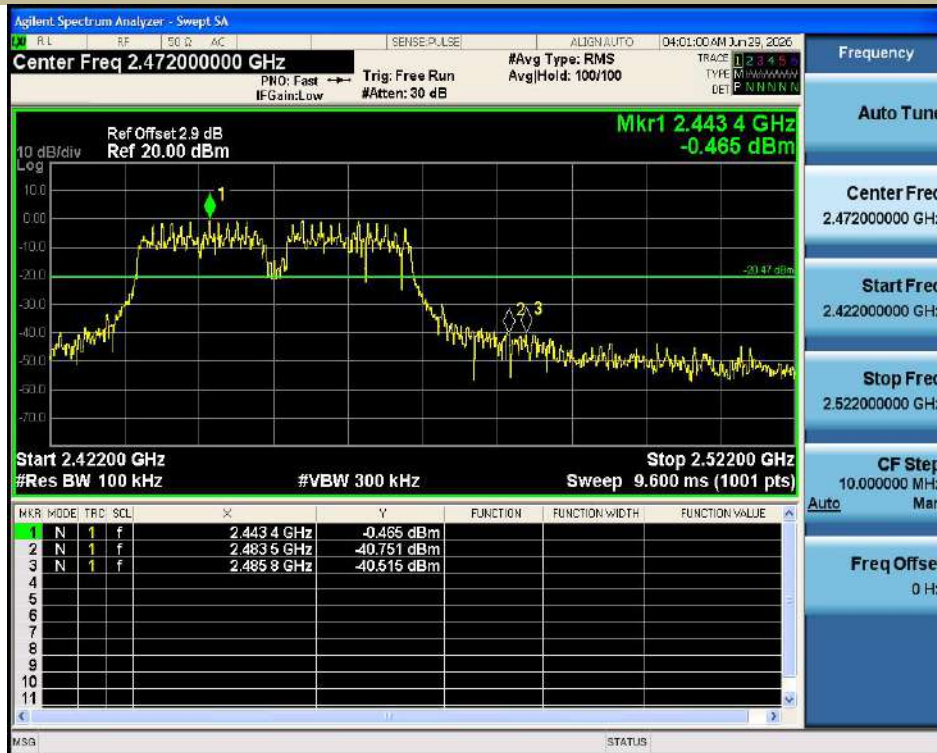
802.11n40



Test Model

Band-edge Conducted Emissions
Channel 9: 2452MHz

802.11n40



6. -6 dB Bandwidth and 99% Bandwidth Measurement

6.1.Limit

Test Requirement: FCC CFR47 Part 15 Section 15.247

Test Method: KDB 558074 D01 15.247 Meas Guidance v05r02; ANSI C63.10-2020+Cor 1-2023

6.2.Test Setup



6.3.Test Procedure

1. Remove the antenna from the EUT and then connect a low RF cable from the antenna port to the spectrum;
2. 6dB: Set RBW = shall be in the range of 1%% to 5% of the OBW but not less than 100 kHz.
Set the VBW $\geq [3 \times \text{RBW}]$.
Detector = peak.
Trace mode = max-hold.
Sweep = No faster than coupled (auto) time.
Allow the trace to stabilize.

99%: Set RBW = 1%~5% of 99% bandwidth, VBW $\geq [3 \times \text{RBW}]$, Sweep = Auto.

6.4.Test Result

Operation Mode	Channel Number	Channel Frequency (MHz)	Measurement Bandwidth (MHz)	Limit (kHz)	Verdict
802.11b	1	2412	9.121	>500	PASS
	6	2437	9.064	>500	PASS
	11	2462	9.588	>500	PASS
802.11g	1	2412	16.43	>500	PASS
	6	2437	16.42	>500	PASS
	11	2462	16.44	>500	PASS
802.11n20	1	2412	17.25	>500	PASS
	6	2437	17.23	>500	PASS
	11	2462	17.07	>500	PASS
802.11n40	1	2422	34.59	>500	PASS
	6	2437	34.72	>500	PASS
	11	2452	35.12	>500	PASS

Test Model

-6dB Bandwidth
Channel 1: 2412MHz

802.11b



Test Model

-6dB Bandwidth
Channel 1: 2437MHz

802.11b



Test Model -6dB Bandwidth
Channel 11: 2462MHz 802.11b



Test Model -6dB Bandwidth
Channel 1: 2412MHz 802.11g



Test Model

-6dB Bandwidth
Channel 6: 2437MHz

802.11g



Test Model

-6dB Bandwidth
Channel 11: 2462MHz

802.11g



Test Model

-6dB Bandwidth
Channel 1: 2412MHz

802.11n20



Test Model

-6dB Bandwidth
Channel 6: 2437MHz

802.11n20



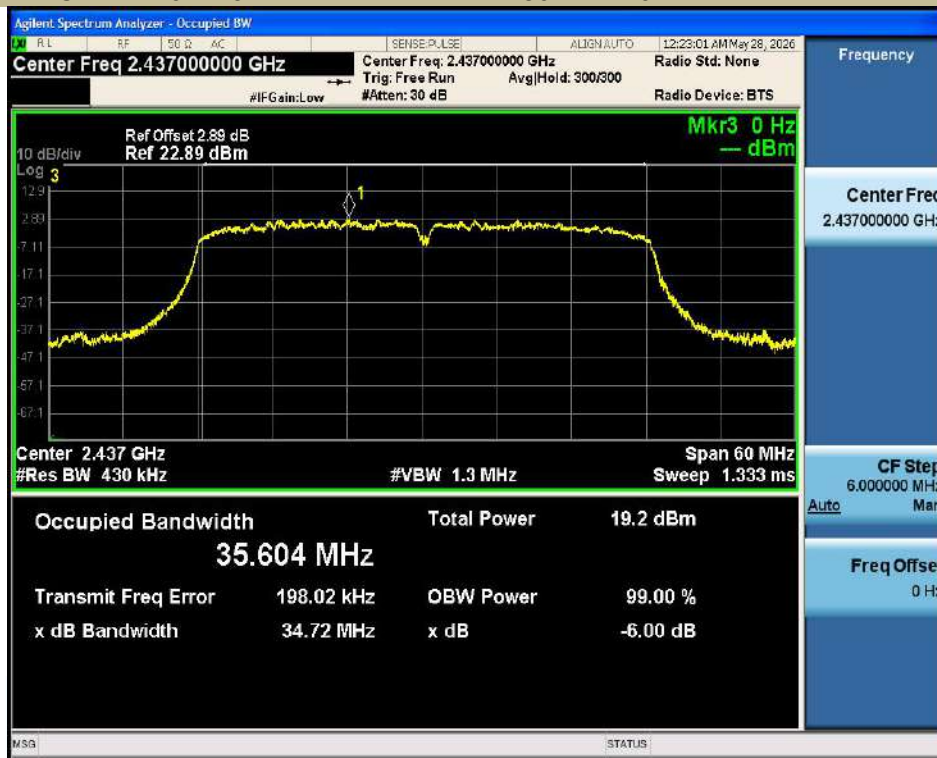
Test Model -6dB Bandwidth
Channel 11: 2462MHz 802.11n20



Test Model -6dB Bandwidth
Channel 3: 2422MHz 802.11n40



Test Model -6dB Bandwidth
Channel 6: 2437MHz 802.11n40



Test Model -6dB Bandwidth
Channel 9: 2452MHz 802.11n40



7. Maximum Output Power& E.I.R.P

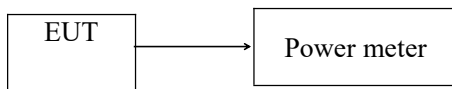
7.1.Limit

Test Requirement: FCC CFR47 Part 15 Section 15.247

Test Method: KDB 558074 D01 15.247 Meas Guidance v05r02;
ANSI C63.10-2020+Cor 1-2023

FCC Part15 (15.247) , Subpart C				
Section	Test Item	Limit	Frequency Range (MHz)	Result
15.247(b)(3)	Output Power	1 watt or 30dBm	2400-2483.5	PASS

7.2.Test Setup



7.3.Test Procedure

1. The maximum peak conducted output power may be measured using a broadband peak RF power meter. The power meter shall have a video bandwidth that is greater than or equal to the DTS bandwidth and shall utilize a fast-responding diode detector.
2. The RF output of EUT was connected to the power meter by RF cable and attenuator. The path loss was compensated to the results for each measurement.
3. Set to the maximum output power setting and enable the EUT transmit continuously.
4. Measure the conducted output power with cable loss and record the results in the test report.
5. Measure and record the results in the report.

7.4.Test Result

Mode	Frequency (MHz)	Conducted Power (dBm)	Limit (dBm)	E.I.R.P value (dBm)	Test Result
b	2412	22.41	30	26.48	PASS
b	2437	22.91	30	26.98	PASS
b	2462	22.57	30	26.64	PASS
g	2412	22.14	30	26.21	PASS
g	2437	22.62	30	26.69	PASS
g	2462	22.21	30	26.28	PASS
n20	2412	21.38	30	25.45	PASS
n20	2437	22.34	30	26.41	PASS
n20	2462	22.06	30	26.13	PASS
n40	2422	21.52	30	25.59	PASS
n40	2437	22.19	30	26.26	PASS
n40	2452	22.04	30	26.11	PASS

Note: EIRP= Total Output Power+ ANT Gain, ANT Gain =4.07dBi.

8. Power Spectral density

8.1.Limit

Test Requirement: FCC CFR47 Part 15 Section 15.247

Test Method: KDB 558074 D01 15.247 Meas Guidance v05r02; ANSI C63.10-2020+Cor 1-2023

FCC Part15 (15.247) , Subpart C				
Section	Test Item	Limit	Frequency Range (MHz)	Result
15.247 (e)	Power Spectral Density	8dBm/3kHz	2400-2483.5	PASS

8.2.Test Setup



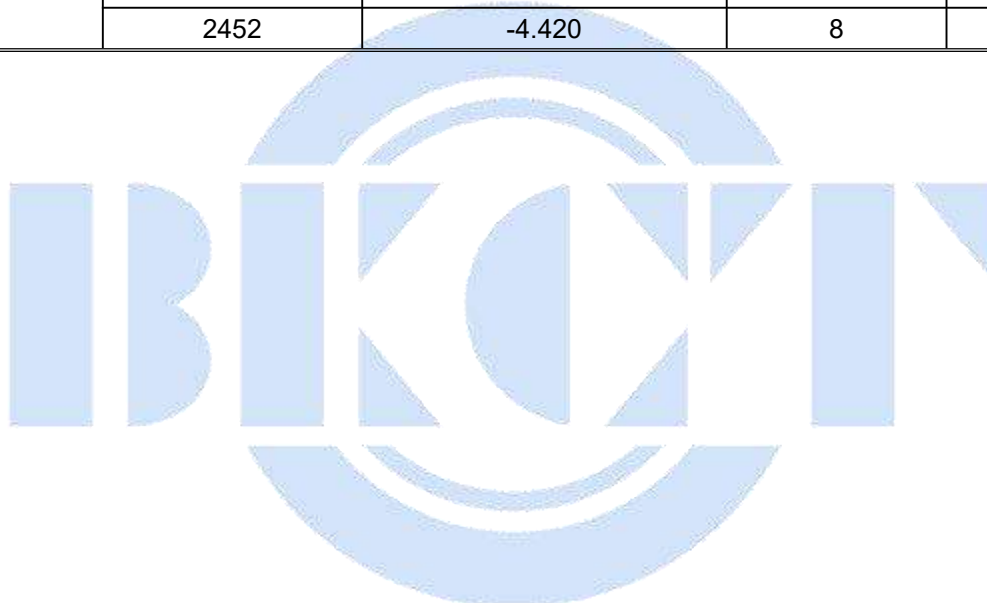
8.3.Test Procedure

KDB 558074 D01 15.247 Meas Guidance v05r02;

1. Remove the antenna from the EUT and then connect a low RF cable from the antenna port to the spectrum.
2. Set instrument center frequency to DTS channel center frequency
3. Set span to > 1.5 times the OBW.
4. Set RBW to: $3 \text{ kHz} \leq \text{RBW} \leq 100 \text{ kHz}$
5. Set VBW $\geq [3 \times \text{RBW}]$
6. Detector = Peak.
7. Trace = Max hold.
8. Sweep time = auto couple.
10. Use the peak marker function to determine the maximum amplitude level.
11. If the measured value exceeds requirement, then reduce RBW (but no less than 3 kHz) and repeat(note that this might require zooming in on the emission of interest and reducing the span to meet the minimum measurement point requirement as the RBW is reduced).
12. Allow the trace to stabilize. Use the marker-delta function to determine the separation between the peaks of the adjacent channels. The limit is specified in one of the subparagraphs of this Section Submit this plot.

8.4. Test Result

Operation Mode	Frequency (MHz)	Power density (dBm/30kHz)	(dBm/3kHz) Limit	Test Result
802.11b	2412	1.415	8	PASS
	2437	1.091	8	PASS
	2462	0.522	8	PASS
802.11g	2412	-0.299	8	PASS
	2437	-0.757	8	PASS
	2462	-1.743	8	PASS
802.11n20	2412	0.157	8	PASS
	2437	-0.597	8	PASS
	2462	-1.488	8	PASS
802.11n40	2422	-3.186	8	PASS
	2437	-3.928	8	PASS
	2452	-4.420	8	PASS



Test Model

Power Spectral Density
Channel 1: 2412MHz

802.11b



Test Model

Power Spectral Density
Channel 6: 2437MHz

802.11b



Test Model

Power Spectral Density
Channel 11: 2462MHz

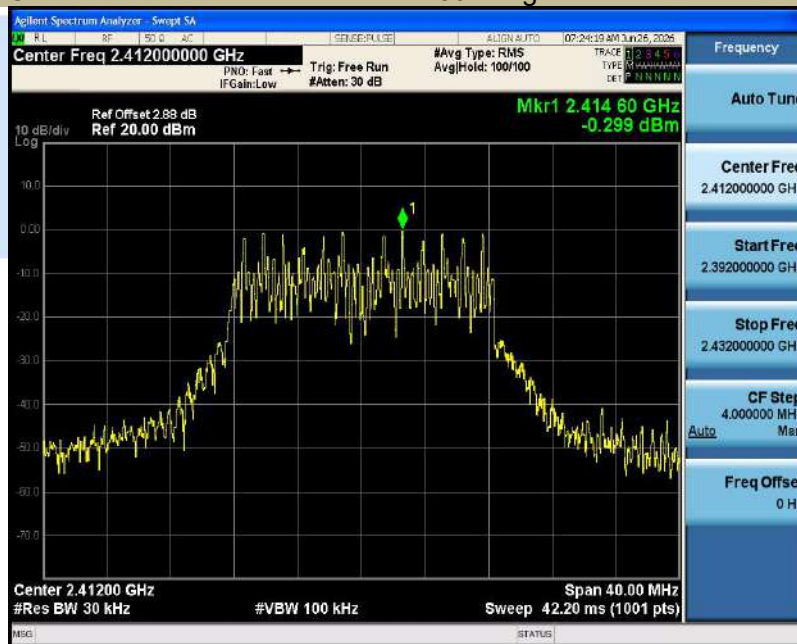
802.11b



Test Model

Power Spectral Density
Channel 1: 2412MHz

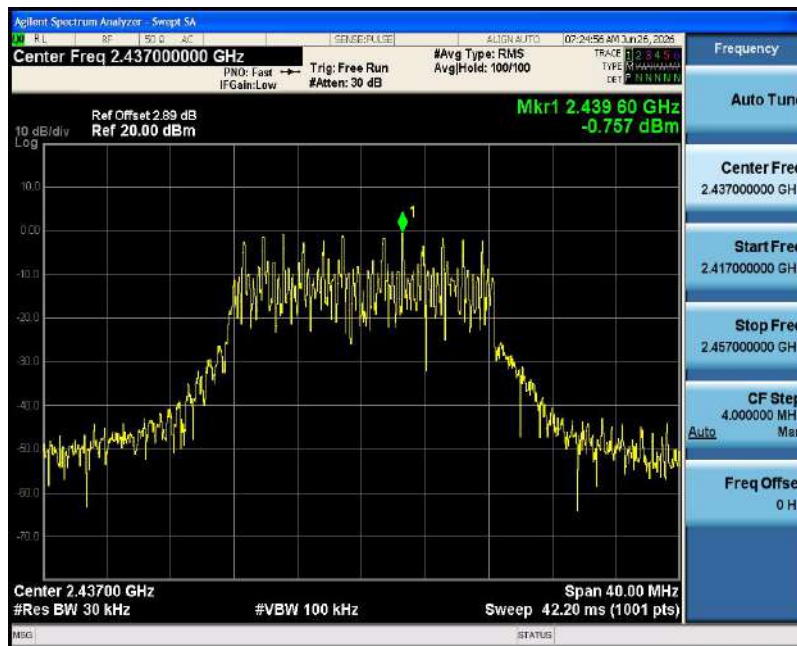
802.11g



Test Model

Power Spectral Density
Channel 6: 2437MHz

802.11g



Test Model

Power Spectral Density
Channel 11: 2462MHz

802.11g



Test Model

Power Spectral Density
Channel 1: 2412MHz

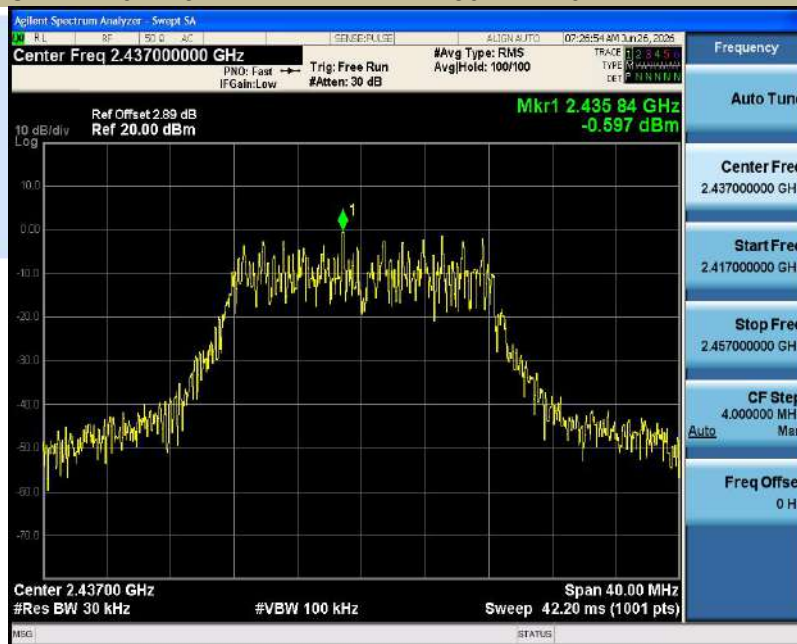
802.11n20



Test Model

Power Spectral Density
Channel 6: 2437MHz

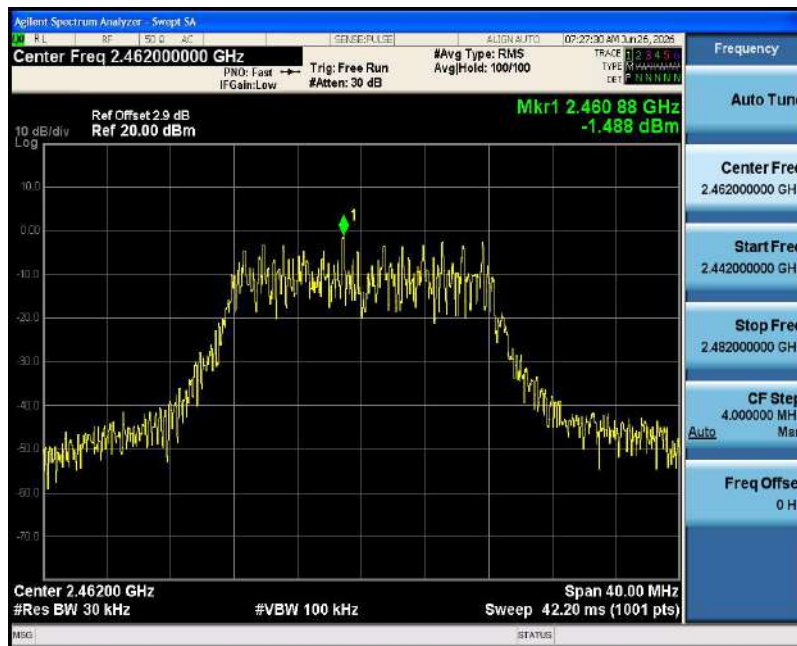
802.11n20



Test Model

Power Spectral Density
Channel 11: 2462MHz

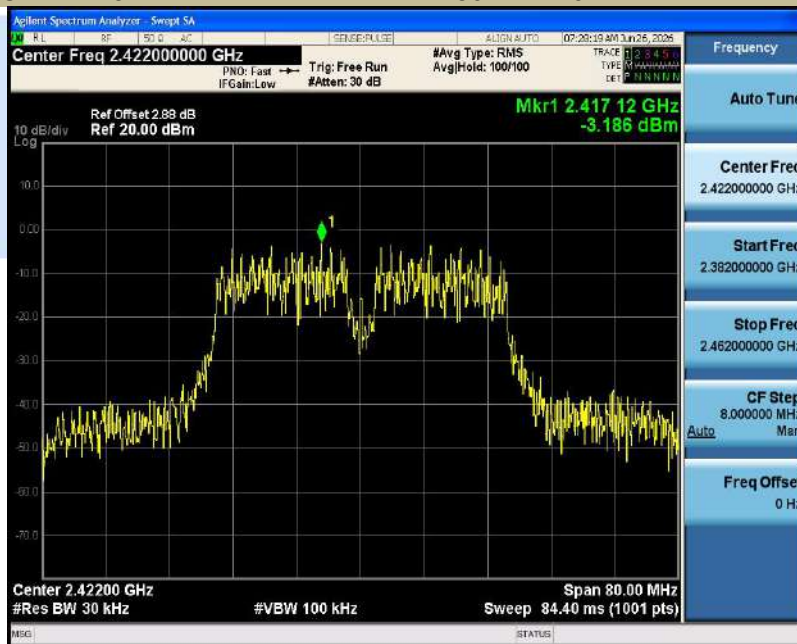
802.11n20



Test Model

Power Spectral Density
Channel 3: 2422MHz

802.11n40



Test Model

Power Spectral Density
Channel 6: 2437MHz

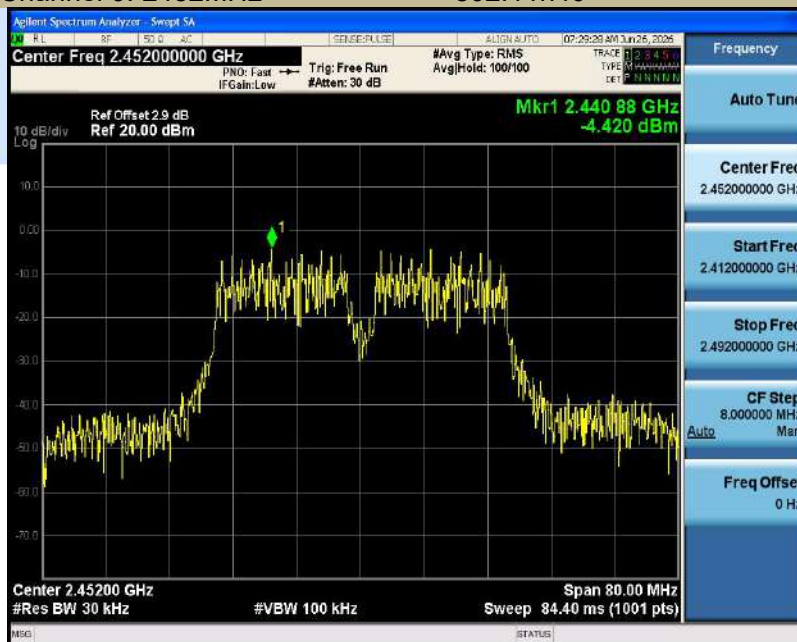
802.11n40



Test Model

Power Spectral Density
Channel 9: 2452MHz

802.11n40



9. Antenna Requirement

According to the FCC Part 15 Paragraph 15.203, an intentional radiator shall be designed to ensure that no antenna other than that furnished by the responsible party shall be used with the device. This product has an ceramic antenna fulfill the requirement of this section.

10. Test Photographs

Please see Attachment for test photographs.

11. Appendix I: Photos of the EUT

12. Please see Attachment for EUT photographs.

